Advanced & Emerging RL Methods

Reinforcement Learning

Advanced & Emerging RL Methods

- Learning with models and fixed data: model-based RL, offline RL
- Learning with demonstrations and humans: behavior cloning, imitation learning, RL from human feedback
- RL as sequence and generative modeling: trajectory modeling, decision transformer, diffusion policies
- Scaling RL to real-world robotics: embodied RL, sim-to-real gap, generalist robot policies, vision-language-action models

Idea of the Chapter

RL approaches that extend or complement standard model-free methods

- model-based techniques
- offline and data-driven approaches
- imitation learning
- generative or multimodal policy methods

Model-Based and Offline RL

Model-Based RL

- Learning dynamics models to plan or improve sample efficiency

Offline RL

- Learning from fixed datasets without online interaction

Learning from Demonstrations

Behavior Cloning

Imitation Learning

- Learning from demonstrations

example: ChatGPT

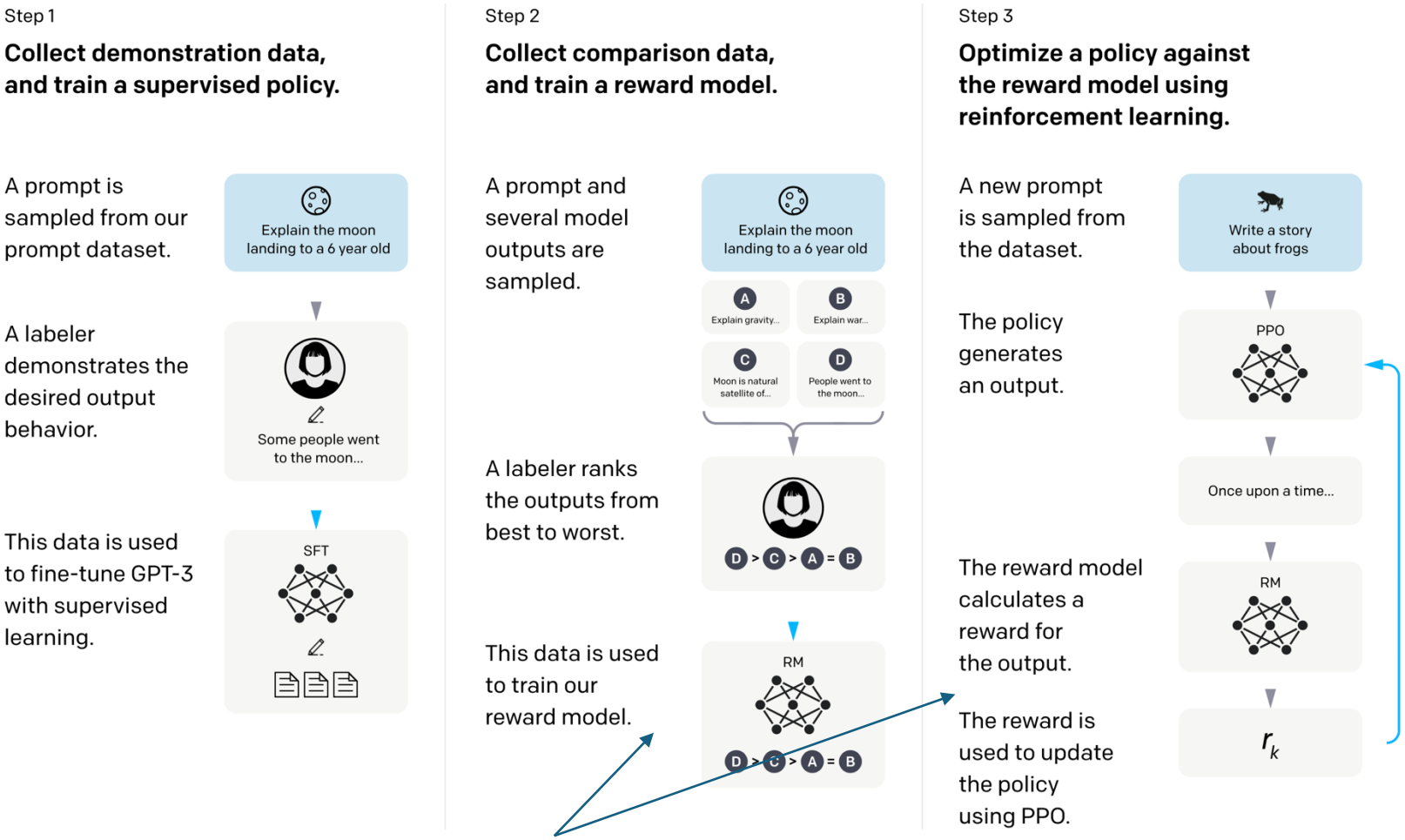
RL from Human Feedback

example for supporting large language models (transformers) with RL

used in famous ChatGPT

goal: improve alignment with user intentions

→ learn from human preferences



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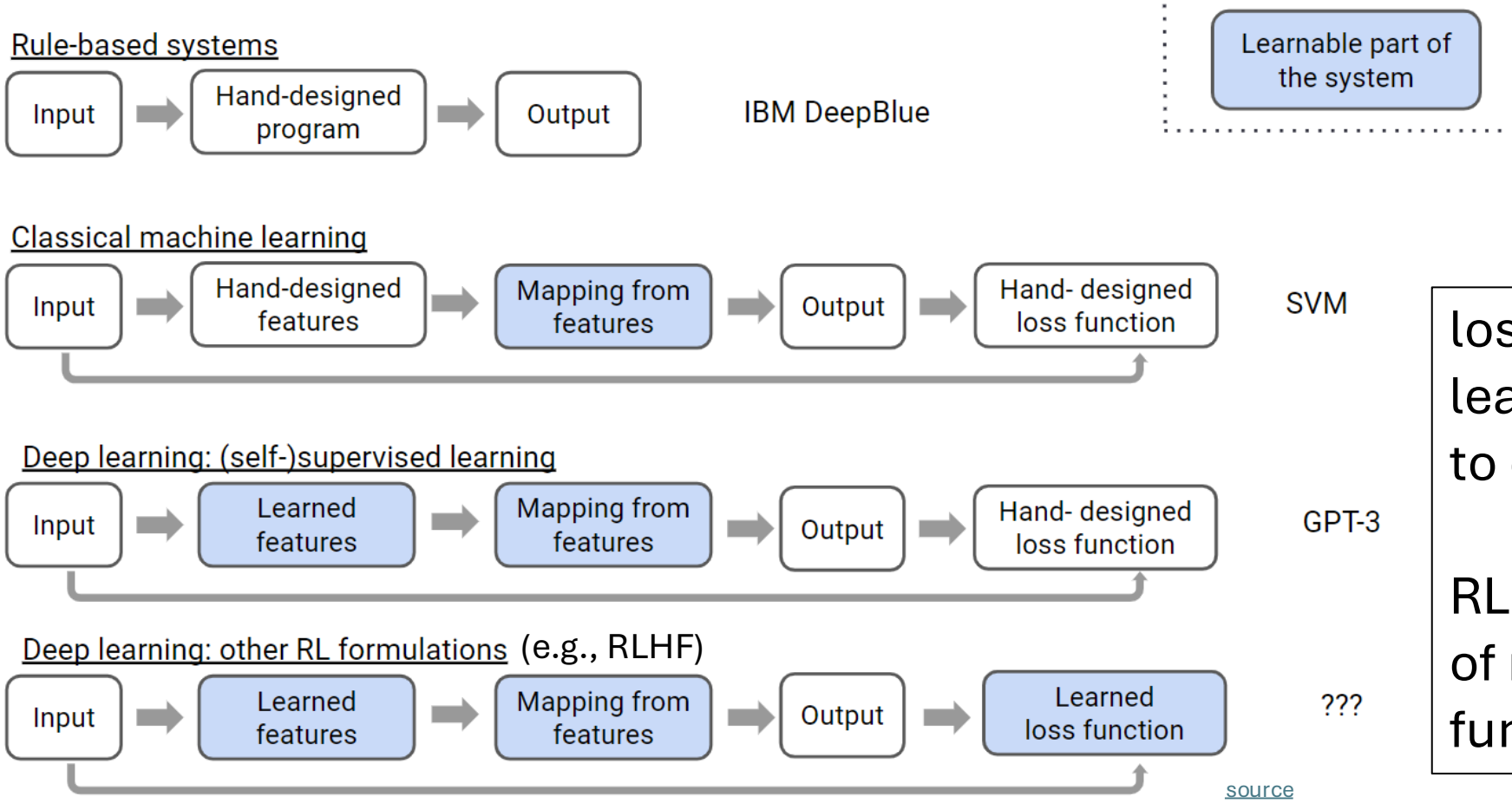
RL looks at reward of text output passages as a whole (rather than token-level loss in supervised learning)

Reward Model



loss function for supervised learning sometimes difficult to define

RL allows learning (instead of manually setting) of loss function

RL as Sequence and Generative Modeling

Trajectory Modeling

- Using sequence models for policy prediction

overcome Markov property

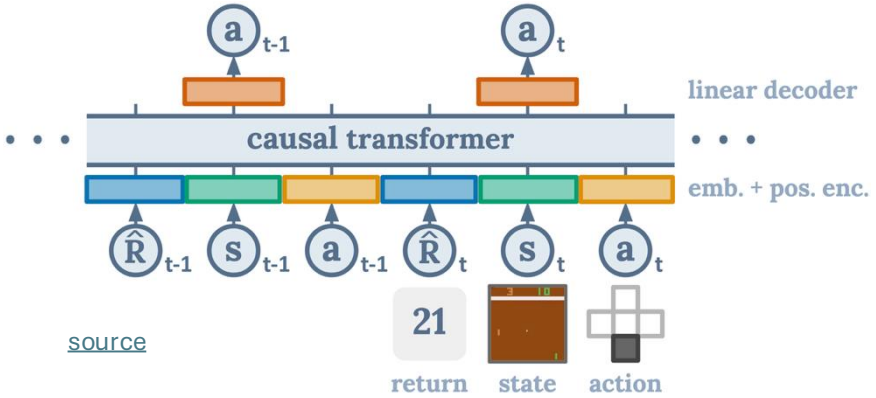
Decision Transformer

generative: transformer decoder architecture to autoregressively model trajectories

credit assignment directly via self-attention: implicitly forming state-return associations via similarity of query and key vectors (maximizing the dot product)

desired return tokens as prompt for action generation

Decision Transformer: conditioning on desired return, past states and actions to generate future actions



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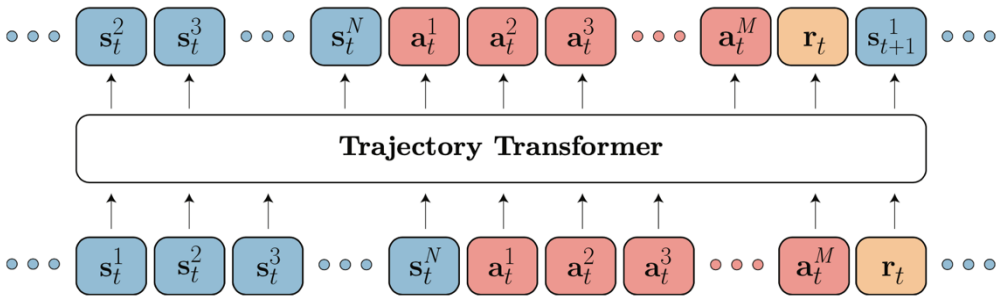
Trajectory Transformer: predicting also states and returns (adding model-based components, planning with beam search)



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Diffusion Policies

- Using generative models to learn flexible policy distributions

Scaling RL to Real-World Robotics

Embodied RL

- Agents interacting with complex, realistic environments

Isaac Sim & Lab

Sim-to-Real Gap

Generalist Robot Policies

- e.g., SmolVLA

Vision-Language-Action Models

Integrating visual and textual inputs for policy learning

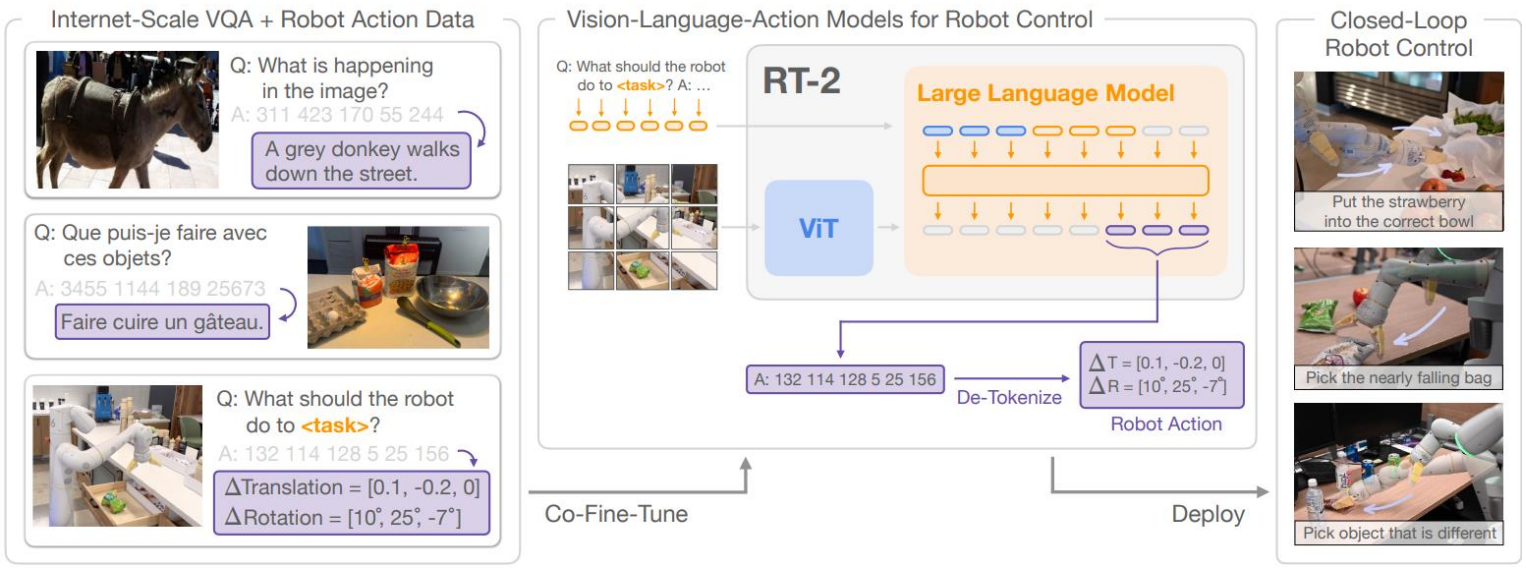
- Vision encoders (ResNet, ViT)
- Language models
- Multimodal fusion
- Tokenization of sensorimotor sequences

can also condition imitation learning
diffusion objective for multimodal outputs

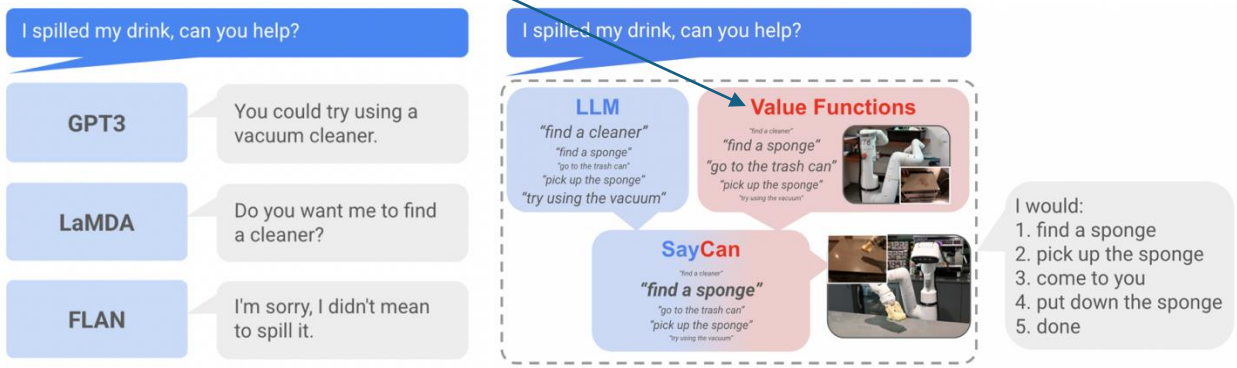
Robotic Control via LLMs (and Vision)

RT-2: vision-language-action model learning from web and robotics data

- representation of actions as tokens
- generalization by using pre-trained vision-language models

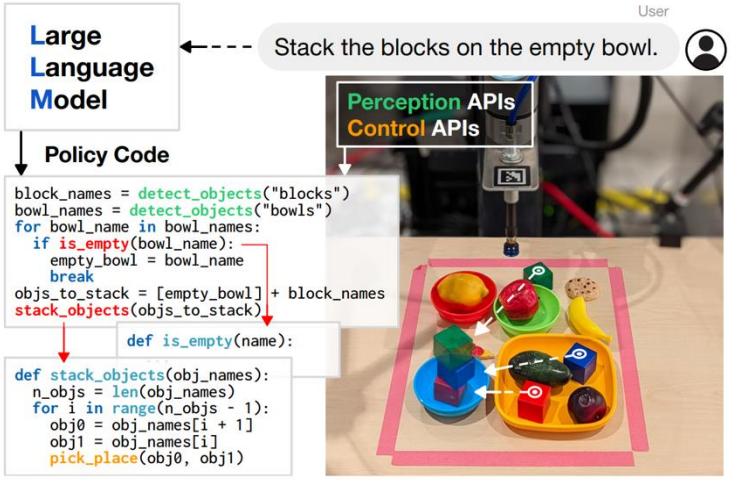


grounding with pre-trained skills (SayCan):



(a) Large Language Models (LLMs) (b) SayCan

Code as Policies:



Practice Project

Gymnasium Pendulum with Decision Transformer